

Notes: Jovanovic and Nyarko (Econometrica, 1996)

Learning by Doing and the Choice of Technology

Contents

1	One-sentence summary	2
2	Big picture	2
2.1	Question	2
2.2	Core answer	2
2.3	Core economic tension	2
2.4	Main result	2
2.5	Main policy implication	2
2.6	Main interpretation	3
3	Incremental contribution of the paper	3
3.1	Learning by doing as both asset and trap	3
3.2	Technology choice under Bayesian learning	3
3.3	Endogenous overtaking	3
3.4	A clean role for transferability	3
3.5	Policy-function characterization	3
3.6	Why the contribution still matters	3
4	Model setup and measurement	4
4.1	Agent, technology grade, and output	4
4.2	Optimal decision and expected output	4
4.3	Technology ladder and knowledge transfer	4
4.4	Variance updating as human-capital dynamics	4
4.5	Myopic technology choice	5
4.6	No-recall constraint	5
5	Model I: learning, transfer, and the option to upgrade	5
5.1	Two forms of learning	5
5.2	The economic meaning of x	5
5.3	The threshold x^*	5

6	Model II: Case A – overtaking and technological lock-in	6
6.1	Case A conditions	6
6.2	Intuition	6
6.3	Policy-function shape	6
6.4	Proposition 3.1: overtaking	6
6.5	Economic message	6
6.6	Competition and spillovers	6
7	Model III: Case B – positive long-run growth	7
7.1	Case B conditions	7
7.2	Intuition	7
7.3	Proposition 4.1: positive long-run growth	7
7.4	Policy-function shape	7
7.5	Economic message	7
8	Comparison of Case A and Case B	7
8.1	The role of experience	7
8.2	What determines the case?	7
8.3	Relation to vintage human-capital models	8
9	Identification-style interpretation	8
9.1	What is being isolated	8
9.2	Key thought experiment	8
9.3	Why the comparison is powerful	8
9.4	Main limitation	8
10	Tables and figures	9
10.1	Notes-created Tables A and B: core objects and parameters	9
10.2	Notes-created Tables C and D: cases and propositions	9
10.3	Single summary table: direction of effects	10
10.4	Extracted Figures 1 and 2	10
10.5	Extracted Figures 3 and 4	11
11	Short summary	12
12	Conclusion	12

12.1 Core conclusion	12
12.2 Economic intuition	12
12.3 Contribution revisited	12
12.4 Implications for growth and development	12
12.5 Implications for organizations	12
12.6 Limitations	12
12.7 Takeaway	13

1 One-sentence summary

Jovanovic and Nyarko build a Bayesian learning-by-doing model in which experience raises productivity on the current technology but transfers imperfectly to better technologies; as a result, accumulated human capital can either promote long-run upgrading or lock skilled agents into old technologies and allow initially less skilled agents to overtake them.

2 Big picture

2.1 Question

How does learning by doing affect the choice of whether to keep exploiting a familiar technology or switch to a better but less familiar technology?

2.2 Core answer

Learning by doing has two opposite effects. It improves decisions on the current technology by reducing uncertainty about that technology, but it can make switching costly because some of the acquired knowledge is technology-specific. The same experience that raises current output may lower the incentive to upgrade.

2.3 Core economic tension

The agent chooses between:

- **Technological deepening:** keep using the current grade and learn its parameter more precisely.
- **Technological opening:** move to a higher grade with greater productive potential but lower expertise.

The paper’s central trade-off is therefore not simply “old technology versus new technology.” It is “known technology with precise beliefs versus better technology with noisier transferable beliefs.”

2.4 Main result

If human capital is sufficiently technology-specific, a highly skilled agent can rationally stay with an old technology forever. A less skilled agent, who has less to lose from switching, may upgrade and eventually overtake the skilled agent. If human capital is sufficiently transferable, experience promotes upgrading and all agents grow in the long run.

2.5 Main policy implication

Policies or institutions that make knowledge more transferable across technologies can matter as much as policies that directly raise the frontier technology. Training systems, modular standards, general-purpose education, and interoperability can reduce the cost of switching to new technologies.

2.6 Main interpretation

The paper provides a formal mechanism for why latecomers can leapfrog incumbents. Incumbents may be trapped by their accumulated expertise in the old technology, while entrants or less experienced agents have weaker attachment to old routines.

3 Incremental contribution of the paper

3.1 Learning by doing as both asset and trap

Earlier learning-by-doing models typically emphasize that experience raises productivity. This paper adds a second channel: experience may reduce adaptability when knowledge is not fully portable across technologies.

3.2 Technology choice under Bayesian learning

The paper models technology adoption as a Bayesian problem. Human capital is summarized by posterior precision about the technology-specific parameter. This makes the adoption decision depend on beliefs, not only on current physical productivity.

3.3 Endogenous overtaking

The paper produces overtaking without assuming that latecomers have better endowments or that leaders face exogenous decline. Overtaking arises endogenously because more expertise on the old technology increases the private cost of switching.

3.4 A clean role for transferability

The parameter α governs how much knowledge transfers across grades of technology. This creates a transparent distinction between general human capital and technology-specific human capital.

3.5 Policy-function characterization

The paper does not only state that switching can be delayed. It characterizes the policy function: in one case, lower uncertainty makes switching less likely; in another case, lower uncertainty makes switching more likely.

3.6 Why the contribution still matters

The mechanism applies to many modern settings: legacy software, production know-how, managerial routines, technology standards, professional skills, platform ecosystems, and organizational learning. Expertise can create inertia.

4 Model setup and measurement

4.1 Agent, technology grade, and output

There is a single risk-neutral agent. At each date t , the agent uses a technology grade n . If the agent chooses decision z , output is

$$q_t = \gamma^n [1 - (y_{nt} - z)^2], \quad \gamma > 1,$$

where

$$y_{nt} = \theta_n + w_{nt}.$$

The unknown term θ_n is the grade-specific parameter. The noise w_{nt} is i.i.d. normal with mean zero and variance σ_w^2 .

4.2 Optimal decision and expected output

Given posterior beliefs, the optimal decision is the posterior mean:

$$z_t = E_t(y_{nt}) = E_t(\theta_n).$$

Expected output is

$$E_t(q_t) = \gamma^n [1 - \text{var}_t(\theta_n) - \sigma_w^2].$$

Interpretation: posterior variance is the state variable for human capital. Lower variance means more expertise and higher expected output on the current grade.

4.3 Technology ladder and knowledge transfer

The relation between the current grade n and a higher grade $n + k$ is

$$\theta_{n+k} = \alpha^{k/2} \theta_n + e_k,$$

where

$$e_k \sim N(0, \rho_k \sigma_e^2), \quad \rho_k = \begin{cases} \frac{1 - \alpha^k}{1 - \alpha}, & \alpha \neq 1, \\ k, & \alpha = 1. \end{cases}$$

If $\alpha = 1$ and $\sigma_e^2 = 0$, knowledge transfers perfectly across grades. If $\alpha = 0$, knowledge is grade-specific.

4.4 Variance updating as human-capital dynamics

Let x denote the posterior variance about the current grade. The paper defines

$$h_1(x) = \frac{\sigma_w^2 x}{\sigma_w^2 + x}$$

for Bayesian updating after using a grade,

$$h_2(x, k) = \alpha^k x + \rho_k \sigma_e^2$$

for transferring knowledge to a grade k steps away, and

$$h(x, k) = h_1(h_2(x, k))$$

for transferring and then updating.

4.5 Myopic technology choice

The agent is myopic and chooses the grade jump $k \in [0, 1]$ that maximizes current expected output. If the status quo grade is n , expected output from a jump of size k is proportional to

$$G(x, k) = \gamma^k \left[1 - \sigma_w^2 - \alpha^k x - \rho_k \sigma_e^2 \right].$$

The term γ^k captures the higher productive potential of the new grade. The term $\alpha^k x + \rho_k \sigma_e^2$ captures the loss of expertise when moving away from the known technology.

4.6 No-recall constraint

Once a grade has been passed over, it is never recalled. In addition, the jump size satisfies $k \leq 1$. This keeps the dynamic problem focused on forward movement along the technology ladder.

5 Model I: learning, transfer, and the option to upgrade

5.1 Two forms of learning

The paper separates learning into two processes:

- **Within-grade learning:** using a technology reveals information about its own parameter and lowers posterior variance.
- **Cross-grade transfer:** knowledge from one grade gives partial information about another grade, but the precision of that information depends on α and σ_e^2 .

5.2 The economic meaning of x

The state variable x has two meanings. A lower x means the agent is more productive today on the current grade. But a lower x also means the agent may have more to lose by leaving the current grade.

5.3 The threshold x^*

The paper defines x^* by the equality

$$G(x^*, 0) = G(x^*, 1).$$

At this threshold, the agent is indifferent between staying at the current grade and making the largest allowed jump. Depending on parameter values, moving below x^* can either encourage or discourage upgrading.

6 Model II: Case A – overtaking and technological lock-in

6.1 Case A conditions

Case A is defined by

$$\alpha\gamma < 1, \quad G(0, 0) > G(0, 1), \quad x^* < x_1^{**},$$

where x_1^{**} is the fixed point of $h(\cdot, 1)$.

6.2 Intuition

The new technology has higher productive potential, but the loss of expertise is sufficiently costly that a highly informed agent prefers to stay. In this case, learning more about the current technology can reduce the willingness to switch.

6.3 Policy-function shape

In Case A:

- low posterior variance means high current expertise;
- high expertise makes staying attractive;
- the agent can become trapped in the current technology;
- a less experienced agent may switch and ultimately grow faster.

6.4 Proposition 3.1: overtaking

If two agents use the same grade but have initial posterior variances $x_0^I < x^* < x_0^{II}$, the initially more skilled agent I has higher expected output at first. But agent I sticks forever, while agent II eventually switches forever. Thus, agent II overtakes agent I in expected output.

6.5 Economic message

Human capital can be a source of inertia. The old expert may rationally refuse to adopt the better technology because the expertise loss from switching is too large.

6.6 Competition and spillovers

The paper argues that Case A can survive competition and informational spillovers. Competition can eventually drive a stagnant old-technology producer out, but it does not guarantee that the incumbent voluntarily switches. Spillovers must arrive early enough; once the leader is far ahead on the old technology, the laggard's information may no longer be strong enough to induce switching.

7 Model III: Case B – positive long-run growth

7.1 Case B conditions

Case B reverses the first two Case A inequalities:

$$\alpha\gamma > 1, \quad G(0,0) < G(0,1).$$

There is no restriction on the relative position of x^* and x_1^{**} .

7.2 Intuition

In Case B, knowledge is sufficiently transferable or the technology ladder is sufficiently productive that experience helps rather than hurts upgrading. As the agent learns, posterior variance falls, and switching to a better grade becomes attractive.

7.3 Proposition 4.1: positive long-run growth

If Case B holds, there exists a positive lower bound on long-run upgrading. Over sufficiently long windows, the agent must choose meaningful upward jumps. Thus, output grows over time from all initial conditions.

7.4 Policy-function shape

The Case B policy function is decreasing in posterior variance. When x is low, the agent is sufficiently skilled to upgrade. When x is high, the agent first stays and learns before upgrading.

7.5 Economic message

When human capital is general enough, learning-by-doing and technological progress reinforce each other. Experience becomes a platform for upgrading rather than an anchor to the past.

8 Comparison of Case A and Case B

8.1 The role of experience

The paper's sharpest contrast is the role of accumulated experience:

- In **Case A**, experience discourages switching and can create stagnation.
- In **Case B**, experience encourages switching and creates sustained upgrading.

8.2 What determines the case?

The most important determinants are:

- α : how transferable knowledge is across technologies;
- γ : how much better higher technology grades are;
- σ_e^2 : how noisy cross-grade transfer is;
- σ_w^2 : how noisy learning is within a grade.

8.3 Relation to vintage human-capital models

The model resembles vintage human-capital models because newer technologies can be more productive. But it adds a distinct mechanism: the agent’s own learning affects whether the move to a new vintage is attractive.

9 Identification-style interpretation

9.1 What is being isolated

This is a theory paper, so there is no empirical identification design. The paper isolates the comparative-static role of knowledge transferability by holding the technology ladder and Bayesian learning structure fixed, then varying the parameters that determine the cost of switching.

9.2 Key thought experiment

Compare two agents at the same technology grade but with different posterior variances. The more skilled agent has lower variance and higher current output. The less skilled agent has higher variance and lower current output. If the first agent sticks and the second switches, the initial ranking can reverse.

9.3 Why the comparison is powerful

The agents are not fundamentally different in ability. The only difference is how much they know about the current technology. Overtaking is therefore caused by endogenous attachment to knowledge, not by superior latent talent of the laggard.

9.4 Main limitation

The model uses a myopic agent and a stylized one-dimensional technology ladder. A fully forward-looking agent, multiple technologies, strategic competition, and richer organizational frictions could alter the boundary between lock-in and growth, but the core mechanism remains clear.

10 Tables and figures

10.1 Notes-created Tables A and B: core objects and parameters

Notes-created Table A: Core objects		Notes-created Table B: Key parameters	
Object	Meaning	Parameter	Economic role
n	Grade of the technology currently used.	α	Transferability of knowledge across grades.
k	Jump size to a higher grade.	σ_e^2	Noise in cross-grade knowledge transfer.
γ^n	Productive potential of grade n .	σ_w^2	Noise in observing the current grade's target.
θ_n	Unknown grade-specific target parameter.	γ	Productivity gain from higher grades.
x	Posterior variance; inverse measure of expertise.	ρ_k	Accumulated uncertainty from jumping k .
$G(x, k)$	Current expected payoff from jumping k .	x^*	Stay-versus-switch threshold.

Read:

- Table A translates the model into a small set of state and control variables.
- Table B highlights that the model turns on the interaction between technology improvement (γ) and knowledge transferability (α).

Economic message: the paper is not about technology progress alone. Growth depends on whether the human capital accumulated on the old technology remains useful after switching.

10.2 Notes-created Tables C and D: cases and propositions

Notes-created Table C: Case comparison			Notes-created Table D: Main propositions	
Case	Condition	Dynamic outcome	Result	Interpretation
Case A	$\alpha\gamma < 1$ and staying is attractive at $x = 0$	Expertise can trap the agent; overtaking is possible.	Proposition 3.1	In Case A, a low-variance expert may stick forever while a higher-variance agent upgrades and overtakes.
Case B	$\alpha\gamma > 1$ and upgrading is attractive at $x = 0$	Experience promotes upgrading; long-run growth occurs.	Proposition 4.1	In Case B, all initial conditions eventually generate positive long-run upgrading.
			Technical lemmas	Establish the shape and monotonicity of policy functions.

Read:

- Case A and Case B reverse the role of experience.
- The same learning technology can generate stagnation or growth depending on transferability.

Economic message: learning by doing is ambiguous for long-run growth. It raises current productivity, but if knowledge is highly specific it can make old routines too attractive.

10.3 Single summary table: direction of effects

Single summary table: comparative-static interpretation

Change	Direct effect	Likely implication
Higher α	More transferability	Easier upgrading; Case B more likely.
Higher γ	Larger frontier gain	Easier upgrading.
Higher σ_ε^2	Noisier transfer	Harder upgrading; lock-in more likely.
Higher σ_w^2	Noisier learning	Slower precision gain.
Lower x	More current expertise	Helps growth in Case B, but can trap in Case A.

Read:

- Most parameters have intuitive monotone effects, but lower x is ambiguous.
- More expertise can either support upgrading or strengthen lock-in.

Economic message: the model's most important insight is the sign reversal on experience: lower uncertainty is not always good for technological progress.

10.4 Extracted Figures 1 and 2

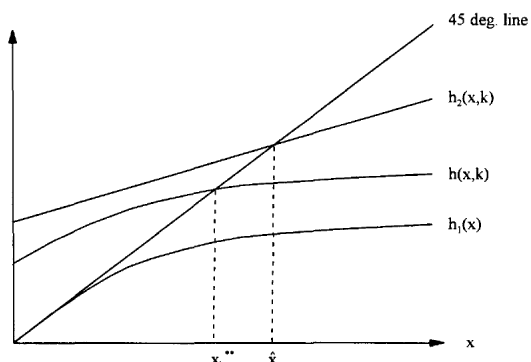


FIGURE 1.—The functions h_1 , h_2 , and h .

Figure 1. The functions h_1 , h_2 , and h .

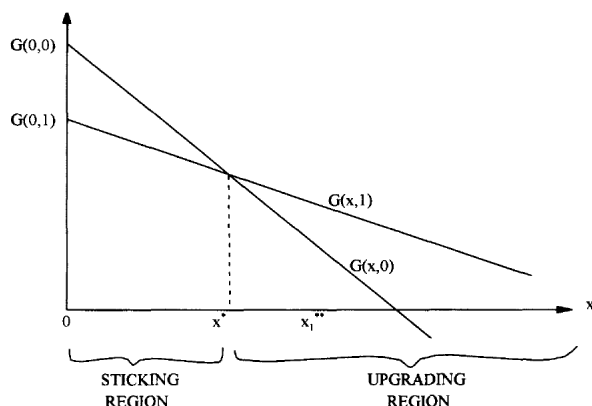


FIGURE 2.—Graph of Case A.

Figure 2. Graph of Case A.

Figure 1 read:

- $h_1(x)$ shows variance after learning within the same grade.
- $h_2(x, k)$ shows variance after transferring knowledge to another grade.
- $h(x, k)$ combines transfer followed by updating.

Figure 1 economic message: using a new technology resets the agent's uncertainty upward through imperfect transfer, but subsequent use lowers uncertainty again. Technology choice is therefore a choice of both productivity level and belief precision.

Figure 2 read:

- In Case A, $G(x, 0)$ and $G(x, 1)$ cross at x^* .
- For low x , the agent stays in the old grade; for high x , upgrading is attractive.
- The skilled agent is precisely the agent most likely to stay.

Figure 2 economic message: a high level of expertise in the current technology can cause rational technological conservatism. This is the graphical core of the overtaking result.

10.5 Extracted Figures 3 and 4

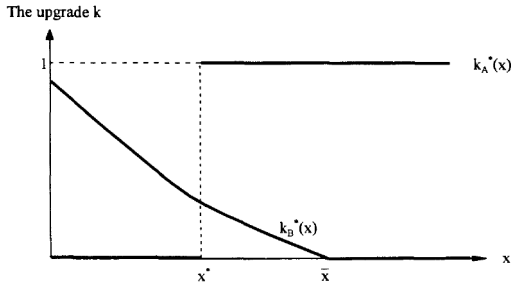


FIGURE 3.—Policy functions in Cases A and B.

Figure 3. Policy functions in Cases A and B.

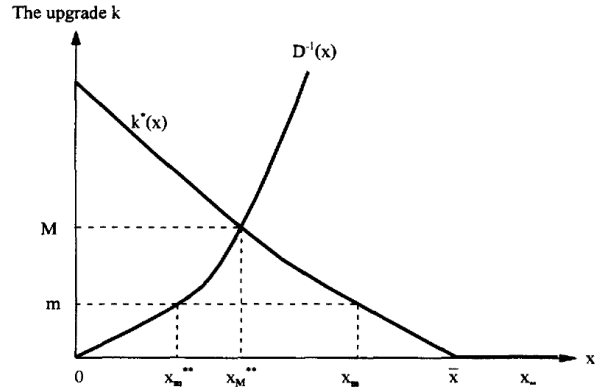


FIGURE 4.—Graph of function D^{-1} and the function $k(x)$.

Figure 4. Graph of D^{-1} and $k(x)$.

Figure 3 read:

- In Case A, the policy jumps only when the agent is sufficiently uncertain about the current grade.
- In Case B, the optimal jump size decreases with x : greater expertise makes upgrading easier.
- The two cases give opposite implications for experience.

Figure 3 economic message: the same learning-by-doing process can produce either lock-in or sustained growth. What matters is whether knowledge transfers well enough to make the next technology worth adopting.

Figure 4 read:

- Figure 4 is part of the technical proof of the policy-function result.
- It shows how the inverse mapping and the candidate jump function intersect.
- The construction supports the monotonicity of the optimal jump rule in Case B.

Figure 4 economic message: the technical appendix backs the economic claim that under Case B, a structured upgrading policy eventually generates repeated movement up the technology ladder.

11 Short summary

The paper studies a single Bayesian agent who learns by doing on a technology ladder. Using a technology lowers uncertainty about that technology, raising expected output. But knowledge transfers imperfectly across grades, so switching to a better technology can reduce expertise. If knowledge is too technology-specific, the skilled agent can become locked into the old technology, while a less skilled agent upgrades and overtakes. If knowledge is sufficiently transferable, experience supports periodic upgrading and long-run growth.

12 Conclusion

12.1 Core conclusion

Learning by doing does not mechanically generate growth. It can deepen expertise on existing technologies, but the same expertise can make switching to new technologies unattractive.

12.2 Economic intuition

The old technology is safe because the agent understands it. The new technology is better but uncertain. The agent upgrades only if the productivity gain from the new grade dominates the expertise loss from moving away from the known grade.

12.3 Contribution revisited

Jovanovic and Nyarko's contribution is to formalize the double-edged nature of human capital. Human capital is both productive knowledge and a form of attachment to a specific technology.

12.4 Implications for growth and development

The model helps explain why leaders may stagnate and latecomers may leapfrog. It also suggests that the economy's ability to grow depends not only on invention, but also on transferability of knowledge across technologies.

12.5 Implications for organizations

Firms with deep legacy expertise may underinvest in new technologies. New entrants, younger workers, or organizations with less entrenched routines may adopt new technologies more readily because they have less specific human capital to lose.

12.6 Limitations

The paper uses a myopic single-agent model and abstracts from many organizational and market frictions. It also focuses on a stylized ladder of technologies. These simplifications make the mechanism transparent but leave room for richer empirical and organizational applications.

12.7 Takeaway

Experience is not always a force for progress. When skills are technology-specific, learning by doing can become learning to stay.